# PROJECT PROPOSAL IN COMPUTER SCIENCE 2

## PROJECT TITLE:

MindType: An Interactive Personality Exploration and Self-Reflection Python Program

## PROBLEM:

Teenage years are an important period of personal development, during which individuals begin to form a stronger understanding of who they are, what they value, how they interact with others, and what they prefer. However, teenagers may sometimes find it difficult to understand their own personality, tendencies, and preferences. This uncertainty can lead them to question how well they know themselves and how they relate to the people around them.

Personality tests are one way people can explore these aspects of themselves. However, some personality assessments may present their results in ways that are difficult for teenagers to understand or may simply provide a personality label without encouraging further reflection. There is therefore an opportunity to create a simple and interactive program that allows teenagers to explore their personality preferences in an accessible and engaging way.

To address this, we propose **MindType**, a personality exploration program inspired by the four personality dimensions associated with the Myers-Briggs Type Indicator (MBTI), developed by the mother-daughter duo, Katharine Cook Briggs and Isabel Briggs Myers, based on the psychological theories of Swiss psychiatrist Carl Jung. It is an independently developed educational program and is not an official MBTI assessment.

The program will present users with a series of age-appropriate questions about their preferences, behaviors, and ways of approaching different situations. These questions will be developed by our group based on the concepts of the MBTI and adapted to include simple, relatable situations that students can easily understand and relate to in their daily lives. Based on their responses, the program will process their answers and determine their preferences across four dimensions: **Extraversion–Introversion (E/I)**, which measures whether a teenager gains energy from socializing or from quiet alone time; **Sensing–Intuition (S/N)**, which looks at whether they process information through practical, concrete facts or through creative, big-picture ideas; **Thinking–Feeling (T/F)**, which determines if they make choices using objective logic or personal values and empathy; and **Judging–Perceiving (J/P)**, which shows whether they prefer a structured, planned lifestyle or a flexible, spontaneous one.

The program will then combine these results to generate a four-letter personality type and provide a brief explanation of the user's identified preferences. Rather than presenting the

result as a fixed definition of who the user is, the program will encourage users to treat the result as a starting point for self-reflection and exploration.

Through this approach, MindType aims to provide teenagers with a simple, interactive, and accessible way to learn more about their personality preferences and reflect on how these tendencies may relate to their everyday experiences and interactions with others.

## QUESTIONNAIRE DEVELOPMENT AND PRETESTING

The questions developed by the project proponents will first be reviewed and pretested with a small group of teenagers to identify potentially confusing, biased, or unclear wording. Feedback from the pretesting will be used to improve the questions before they are incorporated into the program. After the necessary revisions, MindType will undergo user testing with 30 teenagers aged 13 to 19 to evaluate the program's usability and understandability. These respondents will provide feedback on the clarity of the questions, ease of use, and understandability of the generated personality interpretations. The pretesting group will be separate from the 30 respondents participating in the user testing. However, the questions will not be considered a validated psychological assessment, as formal validation of their ability to measure personality traits is beyond the scope of the project.

**SAMPLE QUESTIONS:**

1. Your group is choosing between two ideas for a school project. Both could work, but you can only choose one. What would you focus on more?

A. Which idea is more effective and practical for the project.
B. Which idea would be more meaningful and enjoyable for the group.

A = T
B = F

Why it measures T/F:
Measures whether someone tends to prioritize effectiveness and objective reasoning or personal values and the impact on people.

2. You have several assignments due during the same week. How would you prefer to handle them?

A. Make a plan for when I'll work on each assignment so I know what to expect.
B. Decide what to work on as the week goes on depending on what comes up.

A = J
B = P

Why it measures J/P:
Measures preference for organizing tasks with a plan versus keeping the approach flexible.

3. After a long day of classes at PSHS, you would rather:

A. Hang out and talk with your friends
B. Go somewhere quiet and recharge by yourself

A = E
B = I

Why it measures E/I:
Looks at where you direct your energy and how you recharge.

4. When learning a difficult science topic, you prefer:

A. Specific facts, examples, and details
B. Understanding the bigger idea or concept

A = S
B = N

Why it measures S/N:
Looks at how you process and absorb new information.

## USERS:

The primary target users of MindType are teenagers, particularly those between the ages of **13 and 19**. According to developmental psychologist Erik Erikson (1968), this is a stage in life when individuals face an "identity versus role confusion" crisis as they try to discover who they are. During these years, teenagers may start questioning their personalities, interests, strengths, and preferences. They may also experience changes in their emotions, relationships, and ways of thinking as they grow and become more independent. Because of these changes, some teenagers may feel confused about themselves or find it difficult to understand why they behave or think differently from others. Through MindType, we hope to help them explore their personality preferences in a simple and engaging way. This can encourage them to become more aware of themselves, understand their differences from others, and reflect on how they interact with people in their everyday lives. The questions will be designed to be understandable and relatable to their age group and will focus on everyday situations involving social interaction, decision-making, information processing, and personal preferences.

The program itself will interpret the results based on the answers of users. Each answer choice will be assigned a specific numerical value corresponding to a personality preference. The

program will calculate the scores for each of the four personality dimensions and compare the results to determine which preference is more dominant in each pair. Based on these results, it will generate a four-letter personality type and provide a brief explanation of its possible characteristics. No expert is required to interpret the results because the process will be handled automatically by the program. However, the interpretation will only serve as a guide for self-reflection and will not be considered a professional psychological assessment.

The program is intended to help users explore their personality preferences, recognize possible strengths and tendencies, and reflect on how they interact with other people. It is designed to make personality exploration simple, engaging, and accessible.

Although teenagers are the primary target users, people of other age groups may also use the program if they are interested in exploring their personality preferences.

## OBJECTIVES:

The project aims to:

1. Develop an interactive personality assessment that allows users to explore their personality preferences through a series of age-appropriate questions.
2. Process and analyze users' responses to determine their preferences across four personality dimensions: Extraversion–Introversion, Sensing–Intuition, Thinking–Feeling, and Judging–Perceiving.
3. Generate a four-letter personality type based on the user's calculated preferences.
4. Provide users with a brief and understandable interpretation of their resulting person ality preferences generated by the program.
5. Encourage self-awareness and self-reflection by allowing users to consider how their identified preferences may relate to their everyday behaviors and interactions.
6. Apply computational thinking, modular programming, selection structures, data processing, input validation, and appropriate programming libraries in the development of the program.

## APPLICATION OF PYTHON PROGRAMMING:

Python will be used to develop MindType and implement its main functions. It will process users' responses, calculate scores for the four personality dimensions, determine the resulting four-letter personality type, and generate the corresponding interpretation. Selection structures will be used to determine the dominant preference in each dimension, while modular programming will organize the program into separate functions. Input validation will help ensure that users provide valid responses, and appropriate Python libraries will be used to support the program's functionality.

## FEATURES:

## 1. Introduction and Instructions

The program will begin with an introduction explaining the purpose of MindType and instructions on how to answer the questions. It will also clarify that the result is intended for self-exploration and should not be treated as a definitive description or diagnosis of a person's identity.

## 2. Interactive Personality Questionnaire

Users will answer a series of multiple-choice questions based on common everyday situations and preferences. The questions will be written to be relatable to teenagers.

The questions will assess four personality dimensions:

- **Extraversion (E) / Introversion (I)**
- **Sensing (S) / Intuition (N)**
- **Thinking (T) / Feeling (F)**
- **Judging (J) / Perceiving (P)**

## 3. Input Validation

The program will check whether the user's input is valid before continuing to the next question. Invalid inputs include letters that are not included in the options, numbers, more than one letter, and other elements other than capital or small letter A and B. Invalid inputs will prompt the user to enter an acceptable response again.

This will prevent incorrect or unexpected inputs from affecting the calculated result.

## 4. Response and Score Storage

The program will temporarily store the user's responses and corresponding scores using appropriate variables and data structures.

The stored information will allow the program to process the user's responses after the questionnaire has been completed.

## 5. Personality Score Calculation

The program will count and compare the user's responses for each personality dimension.

For example:

- E compared with I
- S compared with N

- T compared with F
- J compared with P

The higher score in each pair will determine the user's preference for that dimension.

### 6. Personality Type Generation

The program will combine the four determined preferences into a four-letter personality type.

For example:

**E + N + F + P = ENFP**

The program will generate the appropriate combination automatically based on the user's responses.

### 7. Personality Result

After completing the questionnaire, the program will display:

- The user's four-letter personality type
- Their results for each personality dimension
- A brief explanation of their identified preferences
- Possible strengths or tendencies associated with the result
- Reflection prompts that encourage users to think about whether the description relates to their own experience

### 8. Restart or Retake Function

The program may allow users to restart the questionnaire and take the assessment again. This will allow users to explore how different answers can affect their resulting personality preferences.

# INPUT, PROCESSING, AND OUTPUT

### INPUT

The primary inputs of the program will be the user's preferred name or nickname for personalization and answers to the personality questionnaire.

### PROCESSING

The program will:

1. Receive and validate the user's answers.
2. Store the responses.
3. Assign corresponding scores to each personality dimension.
4. Calculate the total score for each side of the four dimensions.
5. Compare the scores within each dimension.
6. Determine the user's four personality preferences.
7. Combine the preferences into a four-letter personality type.
8. Retrieve the corresponding personality description.

### OUTPUT

The program will display:

- The user's personality type
- Their four personality preferences
- A brief personality description
- Possible strengths and tendencies
- Self-reflection prompts

## DATA PROCESSING FLOW

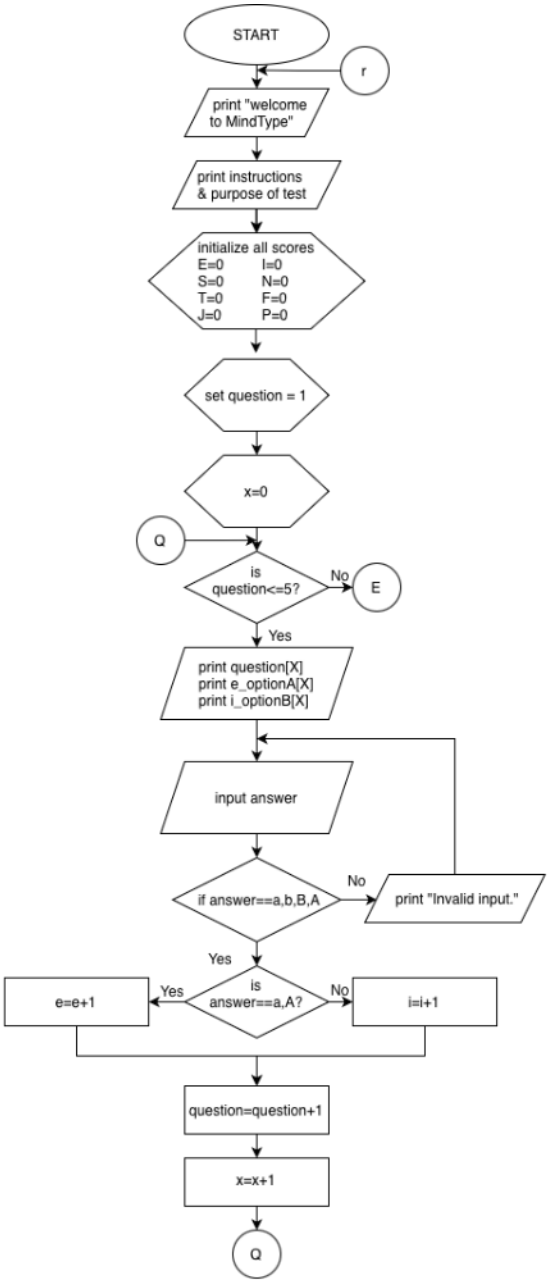
The proposed flow of the program is:

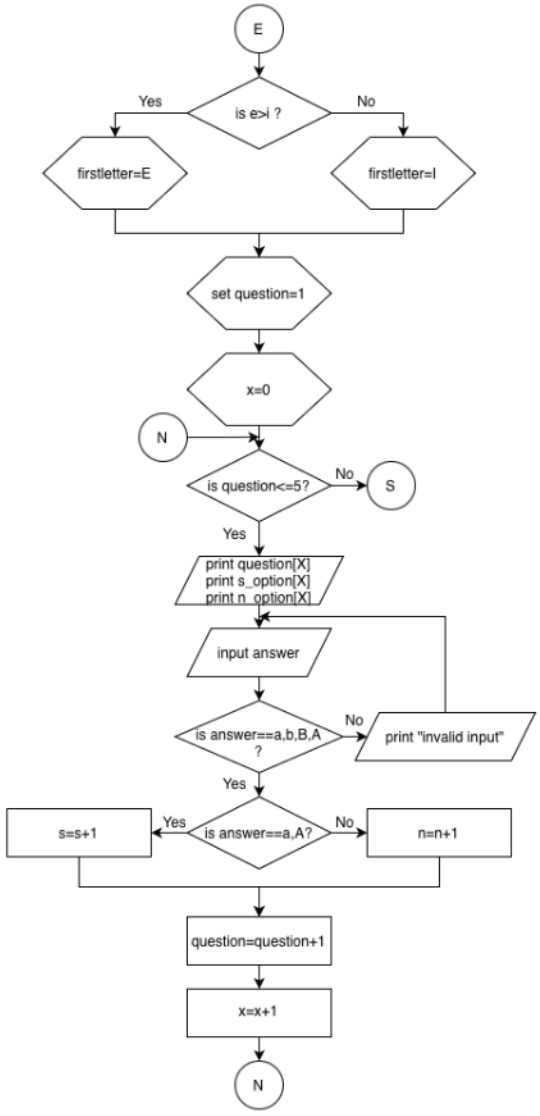
**User → Questions → Answers → Input Validation → Response Storage → Score Calculation → Personality Dimension Comparison → Four-Letter Type → Personality Interpretation → Output**

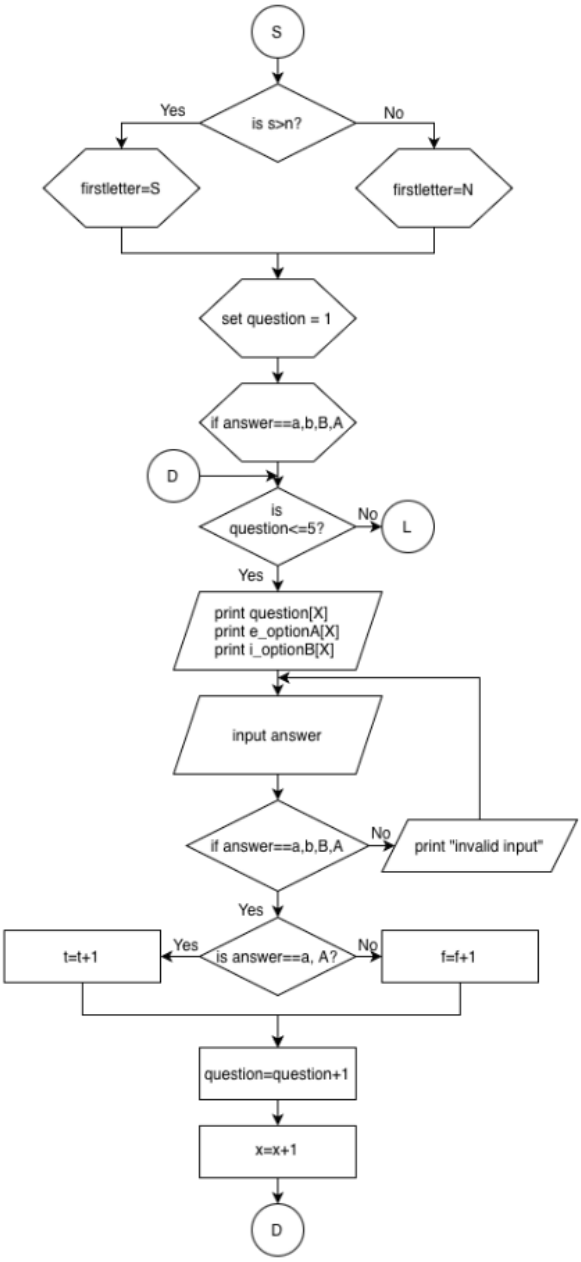
This structure allows the program to demonstrate the data processing steps of collection, input, processing, and output.

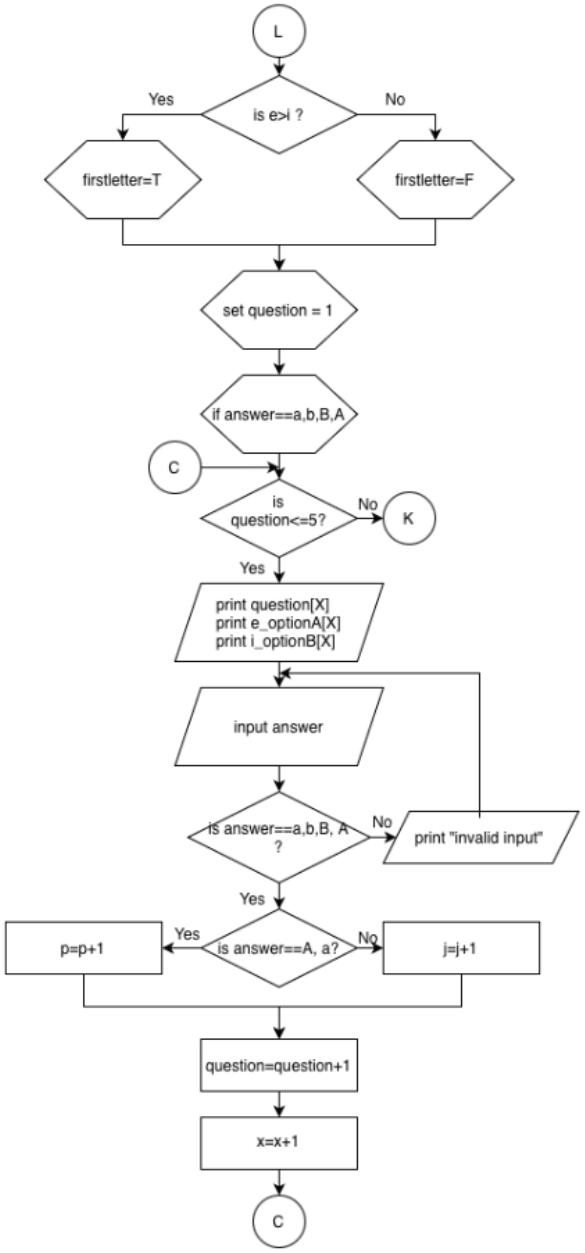
## LOGIC PLAN

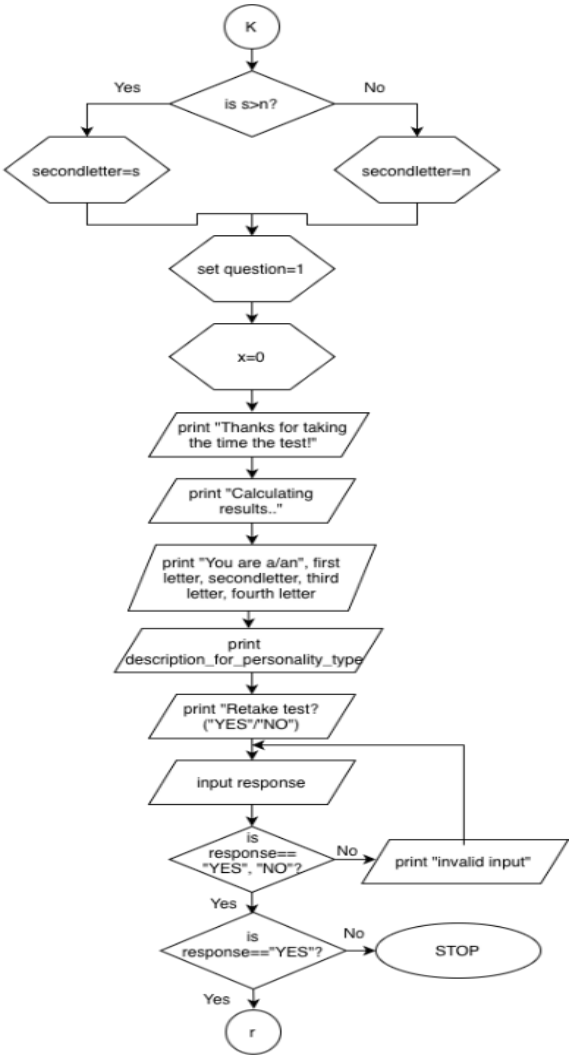
## Flowchart


START
r
print "welcome to MindType"
print instructions & purpose of test
initialize all scores
E=0 I=0
S=0 N=0
T=0 F=0
J=0 P=0
set question = 1
x=0
Q
is question<=5?
No
E
Yes
print question[X]
print e_optionA[X]
print i_optionB[X]
input answer
if answer==a,b,B,A
No
print "Invalid input."
Yes
is answer==a,A?
Yes
e=e+1
No
i=i+1
question=question+1
x=x+1
Q


E
is e>i ?
Yes
No
firstletter=E
firstletter=I
set question=1
x=0
N
is question<=5?
No
S
Yes
print question[X]
print s_option[X]
print n_option[X]
input answer
is answer==a,b,B,A ?
No
print "invalid input"
Yes
is answer==a,A?
Yes
s=s+1
No
n=n+1
question=question+1
x=x+1
N


S
is s>n?
Yes
No
firstletter=S
firstletter=N
set question = 1
if answer==a,b,B,A
D
is question<=5?
No
L
Yes
print question[X]
print e_optionA[X]
print i_optionB[X]
input answer
if answer==a,b,B,A
No
print "invalid input"
Yes
t=t+1
Yes
is answer==a, A?
No
f=f+1
question=question+1
x=x+1
D


L
is e>i ?
Yes
No
firstletter=T
firstletter=F
set question = 1
if answer==a,b,B,A
C
is question<=5?
No
K
Yes
print question[X]
print e_optionA[X]
print i_optionB[X]
input answer
is answer==a,b,B, A ?
No
print "invalid input"
Yes
p=p+1
Yes
is answer==A, a?
No
j=j+1
question=question+1
x=x+1
C




# Pseudocode

Pseudocode:

START

LABEL restart:

```
PRINT "Welcome to MindType"
PRINT instructions and purpose of test


// Step 1: Initialize scores
SET E = 0, I = 0
```

```
SET S = 0, N = 0
SET T = 0, F = 0
SET J = 0, P = 0


// Step 2: Evaluate E / I Trait
SET question = 1
SET X = 0

WHILE question <= 5 DO
    PRINT questions[X]
    PRINT optionA[X]
    PRINT optionB[X]

    // Input validation loop
    REPEAT
        INPUT answer

        IF answer NOT IN ["a", "b", "A", "B"] THEN
            PRINT "Invalid input. Please enter A or B."
        ENDIF

    UNTIL answer IN ["a", "b", "A", "B"]

    IF answer == "a" OR answer == "A" THEN
        SET E = E + 1
    ELSE
        SET I = I + 1
    ENDIF

    SET question = question + 1
    SET X = X + 1
ENDWHILE

IF E > I THEN
    SET firstletter = "E"
ELSE
    SET firstletter = "I"
ENDIF


// Step 3: Evaluate S / N Trait
SET question = 1
SET X = 0

WHILE question <= 5 DO
    PRINT questions[X]
    PRINT optionA[X]
    PRINT optionB[X]

    // Input validation loop
    REPEAT
        INPUT answer

        IF answer NOT IN ["a", "b", "A", "B"] THEN
```

```
            PRINT "Invalid input. Please enter A or B."
        ENDIF

    UNTIL answer IN ["a", "b", "A", "B"]

    IF answer == "a" OR answer == "A" THEN
        SET S = S + 1
    ELSE
        SET N = N + 1
    ENDIF

    SET question = question + 1
    SET X = X + 1
ENDWHILE

IF S > N THEN
    SET secondletter = "S"
ELSE
    SET secondletter = "N"
ENDIF


// Step 4: Evaluate T / F Trait
SET question = 1
SET X = 0

WHILE question <= 5 DO
    PRINT questions[X]
    PRINT optionA[X]
    PRINT optionB[X]

    // Input validation loop
    REPEAT
        INPUT answer

        IF answer NOT IN ["a", "b", "A", "B"] THEN
            PRINT "Invalid input. Please enter A or B."
        ENDIF

    UNTIL answer IN ["a", "b", "A", "B"]

    IF answer == "a" OR answer == "A" THEN
        SET T = T + 1
    ELSE
        SET F = F + 1
    ENDIF

    SET question = question + 1
    SET X = X + 1
ENDWHILE

IF T > F THEN
    SET thirdletter = "T"
ELSE
    SET thirdletter = "F"
```

```
ENDIF


// Step 5: Evaluate J / P Trait
SET question = 1
SET X = 0

WHILE question <= 5 DO
    PRINT questions[X]
    PRINT optionA[X]
    PRINT optionB[X]

    // Input validation loop
    REPEAT
        INPUT answer

        IF answer NOT IN ["a", "b", "A", "B"] THEN
            PRINT "Invalid input. Please enter A or B."
        ENDIF

    UNTIL answer IN ["a", "b", "A", "B"]

    IF answer == "a" OR answer == "A" THEN
        SET J = J + 1
    ELSE
        SET P = P + 1
    ENDIF

    SET question = question + 1
    SET X = X + 1
ENDWHILE

IF J > P THEN
    SET fourthletter = "J"
ELSE
    SET fourthletter = "P"
ENDIF


// Step 6: Output Results
PRINT "Thanks for taking the test!"
PRINT "Calculating results..."

SET personality_type = firstletter + secondletter + thirdletter + fourthletter

PRINT "You are a/an " + personality_type
PRINT description_for_personality_type


// Step 7: Test Retake Decision
PRINT "Retake test? (Yes/No)"

REPEAT
    INPUT response
```

```
    IF response NOT IN ["Yes", "yes", "YES", "No", "no", "NO"] THEN
        PRINT "Invalid input. Please enter Yes or No."
    ENDIF

UNTIL response IN ["Yes", "yes", "YES", "No", "no", "NO"]

IF response == "Yes" OR response == "yes" OR response == "YES" THEN
    GOTO restart
ELSE
    PRINT "Thank you for using MindType!"
    STOP
ENDIF
```

END

# SCOPE AND LIMITATIONS

MindType will focus on personality preferences represented through the four MBTI-inspired dimensions. It will not attempt to diagnose mental health conditions, determine a user's psychological state, or provide professional psychological advice.

The program's results will depend on the user's responses to the questions. Since personality is complex and can vary depending on circumstances, the generated personality type should be treated as an exploratory result rather than a permanent or complete representation of the user.

The initial version of the program will focus on a text-based interactive assessment and its corresponding results.

# DEVELOPMENT CONSIDERATIONS

The project will apply concepts learned in Computer Science 2, including:

- Data collection and processing
- Variables and data structures
- Modular programming using functions
- Selection structures
- Input validation
- String processing
- Appropriate programming libraries and methods
- Debugging and testing
- Code documentation
- Version control using GitHub

The team will also provide appropriate citations for external sources used for the personality descriptions, questions, or other creative and technical content.

## CONCLUSION

MindType is proposed as an interactive personality exploration program designed primarily for teenagers. By combining an accessible questionnaire with automated data processing and personalized results, the program aims to provide users with a simple way to explore their personality preferences and encourage self-awareness.

The project also provides an opportunity to apply the programming concepts and computational thinking skills required in Computer Science 2 through the development of a practical, user-centered program.

## CONTRIBUTORS


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